

Ejercicios punteros

Realizar el seguimiento de los siguientes programas:

Ejercicio 1

```
#include <iostream>
using namespace std;

/* Nota: 'A' es 65 */

int *f1 (int cant)
{
    int *aux = new int[cant];
    return aux;
}

void f2 (int **p, int a)
{
    *p = new int;
    **p = a--;
    cout << **p << (char)a << endl;
}

void f3 (int *&r, int *q)
{
    r = q + 1;
}

int main()
{
    int cant = (udp % 3) + 2; // udp es el ultimo digito de tu DNI
    int *x, *y, *z;
    char a = 'A';
    x = f1(cant);
```

```

    for (int i = 0; i < cant; i++)
        x[i] = a + i;
    f2(&y, a + cant);
    cout << *y << *x << endl;
    f3(z, x);
    cout << *z << endl;
    *z = *x + 2;
    cout << *z << *(x + 1) << *y << endl;
    z = y;
    a = (char)(*y);
    f3(y, x);
    cout << *y << a << *z << x[1] << endl;
    for (int i = 0; i < cant; i++)
        cout << *(x + i) << endl;

    // Instrucciones para liberar la memoria
    return 0;
}

```

Ejercicio 2

```

#include <iostream>
using namespace std;

int* f1(int cant) {
    return new int[cant];
}

void f2 (int* &p, int a) {
    p = new int;
    *p = a++;
    cout << *p << " " << (char)a << endl;
}

int main()

```

```

{
    int udp = 7;
    int cant = (udp % 3) + 2;
    int *x, *y, *z;
    char a = 'A';
    x = f1(cant);
    for (int i=0; i<cant; i++ )
        x[i] = a + i;

    f2( y, a+cant );
    cout << *y << " " << *x << endl;
    z = x++;
    cout << *z << endl;
    *z = *y + 1;
    cout << *z << " " << *(x-1) << " " << *y << endl;
    a = (char)(*y);
    cout << a << " " << *z << " " << *x - 1 << endl;
    for ( int i = 0; i < cant; i++ )
        cout << *(z+i)+i << " ";

    // Liberar memoria
    delete x, y, z;

    return 0;
}

```

Ejercicio 3

```

#include <iostream>
using namespace std;

int* f1(int cant) {
    int *aux = new int[cant];
    return aux;
}

void f2 (int* &p, int a) {

```

```
p = new int;
*p = ++a;
cout << *p << (char)a << endl;
}
```

```
int main()
{
    const int udp = 7;
    int cant = 2*((udp % 3) + 1);
    int *x, *y, *z;
    char a = 'A';
    x = f1( cant );
    for ( int i = 0; i < cant; i++ )
        x[ i ] = a + i;
    f2( y, a + cant );
    cout << *y << *x << endl;
    z = x++;
    cout << *z << endl;
    *z = *y + 1;
    cout << *z << *x << *y << endl;
    a = (char)(*y);
    cout << a << *z << *x - 1 << endl;
    for ( int i = 0; i < cant; i++ )
        cout << *(z + i) ;

    // Liberar memoria

    return 0;
}
```

Ejercicio 4

```
#include <stdlib.h>
#include <iostream>
```

```
using namespace std;

typedef int* Pint;
typedef char* Pchar;

/*
Datos:
'@' es 64
'A' es 65
...
*/

int main(){

    Pint A, C, F;
    Pint* B;
    Pchar D, E;
    char G;
    int H;

    H = 66;
    G = 'D';
    A = new int;
    F = new int;
    (*A) = 64;
    cout << H << G << (*A) << endl;
    B = &F;
    (*F) = (*A) + H - 62;
    (**B) = (*F) - 2;
    cout << (*F) << (*A) << (**B) << endl;
    D = (Pchar)F;
    E = (Pchar)A;
    C = (*B);
    cout << (*D) << (*C) << (*E) << endl;
    (*C) = (*C) - 63;
```

```
if ((*F) == H) {  
    cout << G << H << (*E) << endl;  
}  
  
while ((*C) > 0) {  
    cout << (*E) << (*C) << endl;  
    (*C) = (*C) - 1;  
    (*F) = (*F) - 1;  
    (*A) = 70;  
}  
  
if ((*B) == (*C)) {  
    cout << (*E) << endl;  
}  
  
delete A;  
delete F;  
  
return 0;  
}
```

Ejercicio 5

```
#include <stdlib.h>  
#include <iostream>  
  
using namespace std;  
  
typedef int* Pint;  
typedef char* Pchar;  
  
/*  
Datos:  
'@' es 64  
'A' es 65
```

```
...
*/

int main(){

    Pint A, C, F;
    Pint* B;
    Pchar D, E;
    char G;
    int H;

    H = 70;
    G = 'C';
    F = new int;
    (*F) = H;
    C = new int;
    A = C;
    (*A) = 67;
    H++;

    cout << (*C) << (*A) << (*F) << endl;

    B = &A;
    D = (Pchar)F;
    E = (Pchar)(*B);
    (**B) = (*A) - 63;

    if ( (*E) != G ) {
        cout << (*A) << (*D) << (*C) << endl;
    }

    (*A) = (*A) - (*C) + 66;
    cout << (*E) << (*F) << G << endl;

    while ((*C) > 0) {
```

```
(*E) = 'E';  
(*A) = (*F) - (*C);  
  
cout << (**B) << endl;  
  
(*C)--;  
}  
  
delete A;  
delete F;  
  
return 0;  
}
```